Lecture 18: Finite-Well Resonances and the S-Matrix

2026.11.04

Lecture 17 showed that a finite barrier can transmit perfectly at special energies. We now study positive-energy scattering from an attractive finite well, explain its resonances by adding amplitudes for multiple internal reflections, and organize scattering from either side using the S-matrix.

Questions.

- How can the same well support both bound states and scattering states?
- Why can an attractive well reflect a particle with positive energy?
- Which propagation phase produces a transmission resonance?
- Why must we add amplitudes for different paths before taking probabilities?
- How does the S-matrix map incoming waves to outgoing waves?

Lecture roadmap. We review finite-well bound states and finite-barrier resonances, then change the energy zero to describe the full spectrum of one well. We obtain its positive-energy transmission probability, derive it again by summing internally reflected paths, and finish with the definition and use of the two-channel S-matrix.

Review: bound states and resonant scattering

Finite-well bound states from Lecture 10

With the bottom of a well chosen as the energy zero,

$$V(x) = \begin{cases} 0, & |x| < a/2, \\ V_0, & |x| \geq a/2, \end{cases} \qquad V_0 > 0. \tag{1}$$

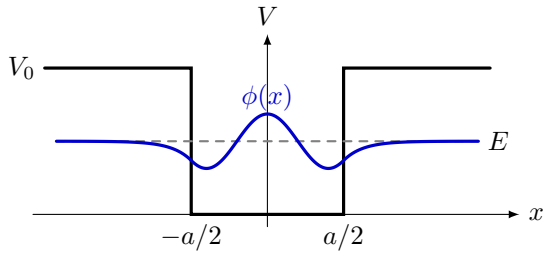


Figure 1: A representative finite-well bound state oscillates inside the well and decays outside. The blue wave function is displaced vertically to the energy line.

A bound state has $0 < E < V_0$. Define

$$k_b = \frac{\sqrt{2mE}}{\hbar}, \qquad \beta = \frac{\sqrt{2m(V_0 - E)}}{\hbar}.$$

The spatial wave function is oscillatory inside and decays outside:

$$\phi(x) = \begin{cases} A_b e^{\beta(x+a/2)}, & x \le -a/2, \\ B_b e^{ik_b x} + C_b e^{-ik_b x}, & |x| < a/2, \\ D_b e^{-\beta(x-a/2)}, & x \ge a/2. \end{cases} \tag{2}$$

Normalizability removes the growing exponential at each infinity. Continuity of ϕ and ϕ' at both boundaries then selects discrete energies. Using parity, the matching conditions become

$$\boxed{\begin{aligned} k_b \tan(k_b a/2) &= \beta \quad \text{(even states)}, \\ -k_b \cot(k_b a/2) &= \beta \quad \text{(odd states)}. \end{aligned}} \tag{3}$$

Only particular energies satisfy these conditions with a nonzero wave function.

Questions.

Question 1: discrete bound-state energies.

Which condition selects the discrete bound-state energies of the finite well?

A) Continuity of ϕ at one boundary only.

B) Decay at both infinities, with ϕ, ϕ' continuous at both boundaries.

C) Equal amplitudes of all plane waves.

D) Unit transmission through the well.

Answer and explanation. Answer: B. A nonzero solution must decay at both infinities and satisfy both matching conditions at both interfaces. These requirements hold only at discrete energies. Perfect transmission instead concerns scattering states.

Finite-barrier resonances from Lecture 17

For the barrier

$$V(x) = \begin{cases} 0, & x < 0, \\ V_0, & 0 < x < L, \\ 0, & x > L, \end{cases} \qquad E > V_0,$$

the exterior and interior wave numbers are

$$k = \frac{\sqrt{2mE}}{\hbar}, \qquad q = \frac{\sqrt{2m(E - V_0)}}{\hbar}.$$

Matching the waves at both interfaces gives

$$T_{\text{barrier}}(E) = \left[1 + \frac{V_0^2}{4E(E - V_0)} \sin^2(qL)\right]^{-1}. \tag{4}$$

Because the exterior wave number is the same on both sides, $R = |B/A|^2$, $T = |F/A|^2$, and $R + T = 1$. In particular,

$$\boxed{qL = n\pi \quad \Longrightarrow \quad T = 1, \qquad R = 0.}$$

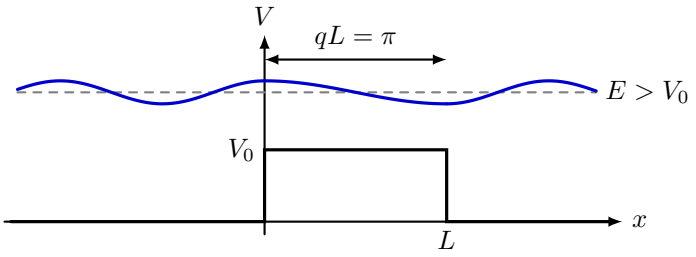


Figure 2: An above-barrier transmission resonance. At $qL = \pi$, an integer number of half-wavelengths fits inside the barrier and the net reflection amplitude vanishes.

This perfect transmission results from interference among reflected waves. We now ask how the result changes when the middle region is a well.

1 One well, two parts of the spectrum

1.1 Changing the energy zero

Subtracting the same constant V_0 from the potential and all energies leaves $E - V(x)$ unchanged:

$$V(x) \longrightarrow V(x) - V_0, \qquad E \longrightarrow E - V_0. \tag{5}$$

The stationary Schrödinger equation therefore has the same spatial solutions and matching conditions. A time-dependent state acquires only a common phase, so probability densities and all physical predictions are unchanged.

Translate the origin to the left edge of the well and write $L = a$. Using V and E for the shifted quantities, the same well is

$$\boxed{V(x) = \begin{cases} 0, & x < 0, \\ -V_0, & 0 < x < L, \\ 0, & x > L, \end{cases} \qquad V_0 > 0.} \tag{6}$$

The old bound-state range $0 < E < V_0$ is now written as

$$\boxed{-V_0 < E < 0.}$$

Negative energy here means energy below the exterior potential. It does not indicate an unphysical state.

Questions.

Question 2: what changes when the energy zero changes?

If the potential and all energies are shifted by the same constant, which statement is correct?

A) Wave functions change; energies do not.

B) Probability densities change.

C) Energy labels change; physical predictions do not.

D) Bound states become scattering states.

Answer and explanation. Answer: C. The spatial eigenfunctions are unchanged because $E - V$ is unchanged. The common time-dependent phase does not affect probabilities. Whether a state is bound depends on its energy relative to the exterior potential, which has shifted by the same amount.

1.2 Bound states and scattering states

For the potential in Eq. (6), the two energy regions have different asymptotic behavior:

energy	outside the well	spectrum
$-V_0 < E < 0$	exponential decay	discrete bound states
$E > 0$	propagating waves	continuous scattering states

(7)

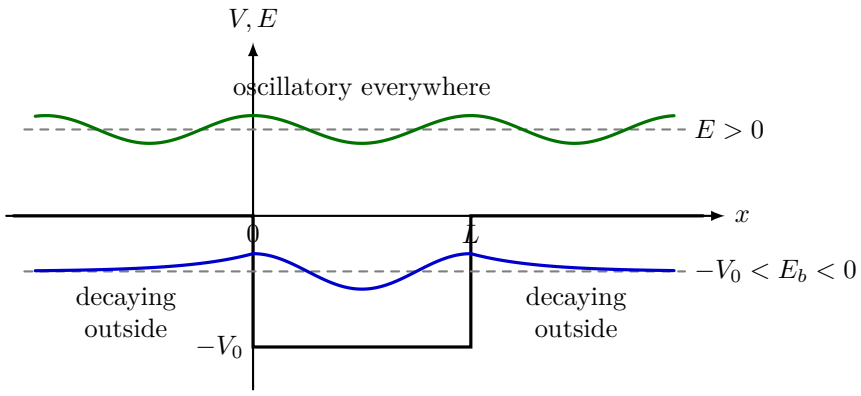


Figure 3: The same finite well has discrete bound states for $-V_0 < E_b < 0$ and a continuous scattering spectrum for $E > 0$. The curves show representative spatial wave profiles displaced vertically to their energy lines.

Bound states must decay at infinity and be normalizable. Scattering eigenfunctions have incoming and outgoing waves at infinity and need not be normalizable individually; localized incident states can be built from them. The same differential equation produces both kinds of solution, with different boundary conditions.

2 Positive-energy scattering from the finite well

2.1 Regional solutions and matching

For $E > 0$, define the exterior and interior wave numbers

$$\boxed{k = \frac{\sqrt{2mE}}{\hbar}, \qquad k_2 = \frac{\sqrt{2m(E+V_0)}}{\hbar}.} \tag{8}$$

The kinetic energy is larger inside the attractive well, so $k_2 > k$ and the interior wavelength $\lambda_2 = 2\pi/k_2$ is shorter.

For incidence from the left,

$$\boxed{\phi(x) = \begin{cases} Ae^{ikx} + Be^{-ikx}, & x < 0, \\ Ce^{ik_2 x} + De^{-ik_2 x}, & 0 < x < L, \\ Fe^{ikx}, & x > L. \end{cases}} \tag{9}$$

Here A is the incident amplitude, B the reflected amplitude, and F the transmitted amplitude. Both directions are needed inside because each interface reflects. There is no left-moving term in the right exterior region because no wave is incident from $+\infty$.

Continuity of ϕ and ϕ' gives

$$\begin{aligned} A + B &= C + D, & k(A - B) &= k_2(C - D), \\ Ce^{ik_2 L} + De^{-ik_2 L} &= Fe^{ikL}, & k_2(Ce^{ik_2 L} - De^{-ik_2 L}) &= kFe^{ikL}. \end{aligned} \tag{10}$$

These are the same matching equations as for the barrier, with the appropriate interior wave number. Replacing the middle-region potential $+V_0$ by $-V_0$ in the barrier result gives

$$\boxed{T(E) = \left[1 + \frac{V_0^2}{4E(E+V_0)} \sin^2(k_2 L)\right]^{-1}.} \tag{11}$$

2.2 Probabilities are ratios of currents

For a stationary state $\psi(x,t) = \phi(x)e^{-iEt/\hbar}$, the probability current can be calculated from ϕ:

$$j = \frac{\hbar}{2mi}\left(\phi^* \phi' - \phi \phi'^*\right).$$

In the exterior regions,

$$j_{\text{in}} = \frac{\hbar k}{m}|A|^2, \qquad j_{\text{refl}} = -\frac{\hbar k}{m}|B|^2, \qquad j_{\text{trans}} = \frac{\hbar k}{m}|F|^2. \tag{12}$$

Hence

$$\boxed{R = \frac{|j_{\text{refl}}|}{j_{\text{in}}} = \left|\frac{B}{A}\right|^2, \qquad T = \frac{j_{\text{trans}}}{j_{\text{in}}} = \left|\frac{F}{A}\right|^2, \qquad R + T = 1.} \tag{13}$$

The amplitude ratios give the probabilities directly here because the exterior wave numbers, and therefore the velocity factors, are equal. At an individual interface with unequal wave numbers, the velocity factor must be retained.

2.3 Transmission resonances

Equation (11) gives

$$\boxed{k_2 L = n\pi \quad \Longrightarrow \quad T = 1, \qquad R = 0.} \tag{14}$$

An integer number of interior half-wavelengths fits across the well: $L = n\lambda_2/2$. Only integers giving $E > 0$ belong to the scattering resonances considered here. Although both interfaces individually reflect, the net reflected amplitude vanishes at resonance.

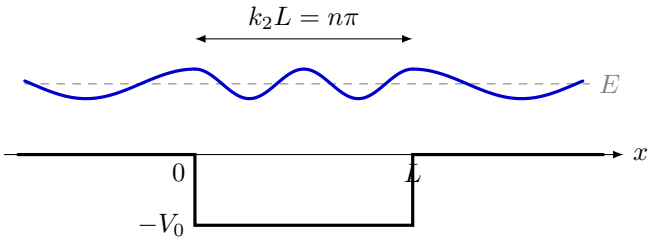


Figure 4: A resonant wave has a shorter wavelength inside the well. The blue curve shows a spatial wave profile displaced vertically to the energy line; it is not a classical trajectory.

Questions.

Question 3: identify the resonance.

Which condition gives a positive-energy transmission resonance of the well?

A) $kL = n\pi$

B) $k_2 L = n\pi$

C) $\kappa L = n\pi$

D) $E = -V_0$

Answer and explanation. Answer: B. The relevant propagation occurs inside the well, so its phase involves k_2, not the exterior wave number k. When $k_2 L = n\pi$, the sine term in Eq. (11) vanishes and $T = 1$.

3 Why the resonance occurs: multiple internal reflections

3.1 Add amplitudes for indistinguishable paths

One might try to multiply the transmission probabilities of a downward step and an upward step:

$$T_{\text{well}} \stackrel{?}{=} T_{\text{down step}} T_{\text{up step}}.$$

This includes only a direct crossing and has no dependence on L. A wave can also make one, two, or arbitrarily many round trips before leaving. When these

alternatives are coherent and no path information distinguishes them, their complex amplitudes must be added.

The same rule produces double-slit interference:

$$\psi = \psi_1 + \psi_2, \qquad |\psi|^2 = |\psi_1|^2 + |\psi_2|^2 + 2\,\mathrm{Re}(\psi_1^* \psi_2). \tag{15}$$

Adding probabilities alone would lose the relative phases and the interference term.

Questions.

Question 4: what must be added?

How should the direct path and paths with internal round trips be combined?

A) Add probabilities.

B) Keep only the direct path.

C) Add amplitudes, then take the modulus squared.

D) Discard all relative phases.

Answer and explanation. Answer: C. As in the double-slit experiment, indistinguishable alternatives contribute amplitudes. Relative phases determine whether they reinforce or cancel, so taking modulus squares before summing would miss the resonance.

3.2 Single-interface amplitudes

Place one interface at $x = 0$. A unit-amplitude wave incident from a region with wave number $k_i > 0$ into one with wave number $k_j > 0$ has

$$\phi_i = e^{ik_i x} + r_{ij} e^{-ik_i x}, \qquad \phi_j = t_{ij} e^{ik_j x}.$$

Matching ϕ and ϕ' gives

$$1 + r_{ij} = t_{ij}, \qquad k_i(1 - r_{ij}) = k_j t_{ij},$$

and therefore

$$\boxed{r_{ij} = \frac{k_i - k_j}{k_i + k_j}, \qquad t_{ij} = \frac{2k_i}{k_i + k_j}.} \tag{16}$$

These are amplitudes, not probabilities. For this interface,

$$R_{ij} = |r_{ij}|^2, \qquad T_{ij} = \frac{k_j}{k_i} |t_{ij}|^2.$$

The indices specify the incident and final regions; for incidence in the opposite direction, use local coordinates along the incident direction.

For the well, label the exterior regions 1 and 3 and the interior region 2. Then $k_1 = k_3 = k$, and

$$t_{12} = \frac{2k}{k + k_2}, \qquad t_{23} = \frac{2k_2}{k + k_2}, \tag{17}$$

while the two reflection amplitudes for waves incident from inside are

$$\boxed{r_{21} = r_{23} \equiv r = \frac{k_2 - k}{k_2 + k}.} \tag{18}$$

3.3 Direct transmission and one round trip

Measure the transmitted amplitude at the right interface relative to the incident amplitude at the left interface. The direct path enters the well, propagates through L, and exits:

$$\boxed{\mathcal{A}_0 = t_{12} e^{ik_2 L} t_{23}.} \tag{19}$$

Since $|e^{ik_2 L}| = 1$, propagation changes the phase but not the modulus.

Why both traversals contribute $e^{ik_2 L}$.
For a right-moving wave inside the well,

$$\phi_{\rightarrow}(x) = a_0 e^{ik_2 x}.$$

Starting at $x = 0$, its amplitude at $x = L$ is therefore

$$\phi_{\rightarrow}(L) = \phi_{\rightarrow}(0) e^{ik_2 L}.$$

The return wave moves to the left and has the spatial dependence $e^{-ik_2 x}$. If its amplitude immediately after reflection at $x = L$ is a_L, it is most clearly written relative to that starting point:

$$\phi_{\leftarrow}(x) = a_L e^{-ik_2 (x - L)}.$$

When it reaches $x = 0$,

$$\phi_{\leftarrow}(0) = a_L e^{-ik_2 (0 - L)} = a_L e^{ik_2 L}.$$

Thus the return trip also contributes $e^{ik_2 L}$, and a complete round trip contributes

$$\boxed{e^{ik_2 L} e^{ik_2 L} = e^{2ik_2 L}, \qquad \text{not generally 1.}}$$

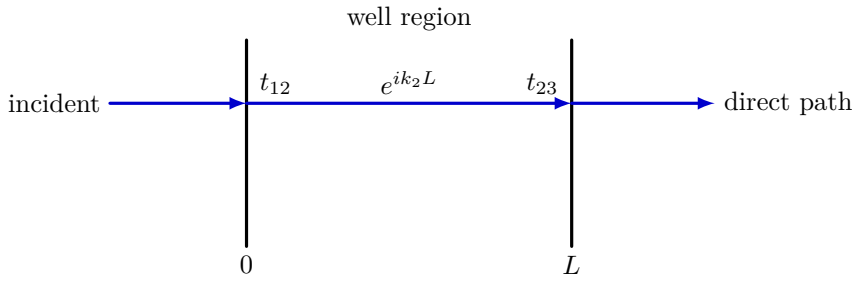


Figure 5: The direct transmitted path enters the well, propagates once across its width, and exits.

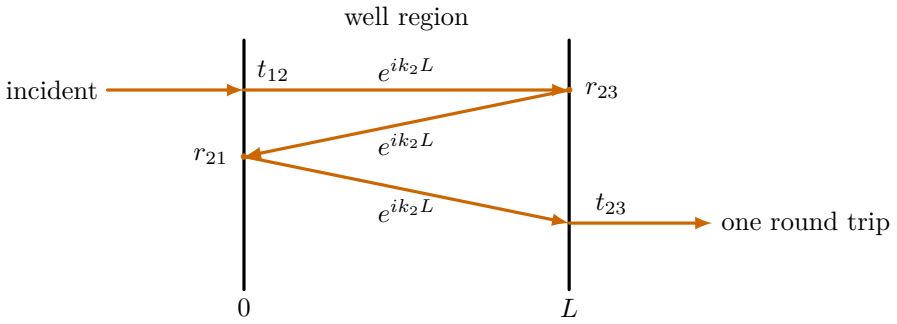


Figure 6: A path with one internal round trip has two reflection amplitudes and an additional propagation distance $2L$. Vertical offsets only separate the path segments visually.

A path with one extra round trip reflects first at $x = L$, then at $x = 0$, and propagates an extra distance $2L$:

$$\begin{aligned}\mathcal{A}_1 &= t_{12}e^{ik_2L}r_{23}e^{ik_2L}r_{21}e^{ik_2L}t_{23} \\ &= \mathcal{A}_0\left(r_{21}r_{23}e^{2ik_2L}\right). \end{aligned} \tag{20}$$

Every additional round trip supplies the same factor, so

$$\boxed{\mathcal{A}_n = t_{12}t_{23}e^{ik_2L}\left(r_{21}r_{23}e^{2ik_2L}\right)^n, \qquad n = 0, 1, 2, \ldots.} \tag{21}$$

The factor e^{ik_2L} is acquired on either traversal: it describes propagation along the wave's direction of travel over a distance L.

3.4 Summing the geometric series

Let $z = r_{21}r_{23}e^{2ik_2L} = r^2e^{2ik_2L}$. Since $k, k_2 > 0$, $|r| < 1$ and $|z| = r^2 < 1$. Thus

$$\sum_{n=0}^{N} z^n = \frac{1 - z^{N+1}}{1 - z} \quad \longrightarrow \quad \sum_{n=0}^{\infty} z^n = \frac{1}{1 - z}.$$

Adding all transmitted amplitudes gives

$$\boxed{\widetilde{t} \equiv \sum_{n=0}^{\infty} \mathcal{A}_n = \frac{t_{12}t_{23}e^{ik_2L}}{1 - r^2e^{2ik_2L}}.} \tag{22}$$

Here $\widetilde{t}$ uses the two interface positions as its phase reference. In the global plane-wave convention of Eq. (9),

$$\widetilde{t} = \frac{Fe^{ikL}}{A}, \qquad t_L \equiv \frac{F}{A} = e^{-ikL}\widetilde{t}.$$

The extra phase has unit modulus, so both give the same transmission probability $T = |\widetilde{t}|^2 = |t_L|^2$.

Now

$$t_{12}t_{23} = \frac{4kk_2}{(k+k_2)^2} = 1 - r^2,$$

and

$$\begin{aligned} |1 - r^2 e^{2ik_2 L}|^2 &= 1 + r^4 - 2r^2 \cos(2k_2 L) \\ &= (1-r^2)^2 + 4r^2 \sin^2(k_2 L). \end{aligned} \tag{23}$$

Consequently,

$$T = \left[1 + \frac{4r^2}{(1-r^2)^2} \sin^2(k_2 L)\right]^{-1}. \tag{24}$$

Using the definitions of k and k_2,

$$\frac{4r^2}{(1-r^2)^2} = \frac{(k_2^2 - k^2)^2}{4k^2 k_2^2} = \frac{V_0^2}{4E(E+V_0)}, \tag{25}$$

which reproduces Eq. (11). Boundary matching and the path sum describe the same coherent scattering process.

At resonance, $e^{2ik_2 L} = 1$, so the denominator of Eq. (22) is $1 - r^2$, which cancels the entry and exit amplitude product. Therefore $|\widetilde{t}| = 1$ and $T = 1$, even though either interface alone reflects.

Questions.

Question 5: one additional round trip.

One additional round trip inside the well multiplies a path amplitude by:

A) $r_{23}e^{ik_2 L}$

B) $r_{21}r_{23}e^{ik_2 L}$

C) $|r_{21}r_{23}|^2$

D) $r_{21}r_{23}e^{2ik_2 L}$

Answer and explanation. Answer: D. A round trip includes two reflections and an extra propagation distance $2L$. Both reflection amplitudes and the full complex propagation phase must be retained before the paths are summed.

4 The S-matrix: organizing all scattering amplitudes

4.1 Incoming and outgoing channels

So far we have imposed incidence from the left. More generally, waves can arrive from either side. For a localized potential with the same exterior potential on both sides, write the asymptotic stationary solution at $E > 0$ as

$$\phi(x) \sim \begin{cases} Ae^{ikx} + Be^{-ikx}, & x \to -\infty, \\ Ce^{ikx} + De^{-ikx}, & x \to +\infty. \end{cases} \tag{26}$$

The letters C, D now denote *right-exterior* coefficients, rather than the interior coefficients used earlier.

For the time dependence $e^{-iEt/\hbar}$ and $k > 0$, e^{ikx} travels to the right and e^{-ikx} to the left. A wave is incoming when it travels toward the potential, so

incoming amplitudes: A, D, outgoing amplitudes: B, C.

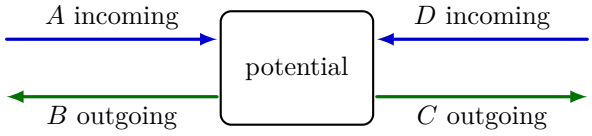


Figure 7: Incoming and outgoing channels are identified by propagation direction relative to the potential.

4.2 Definition and the two columns

Linearity of the Schrödinger equation gives a linear map from the incoming amplitudes to the outgoing amplitudes:

$$\boxed{\begin{pmatrix} B \\ C \end{pmatrix} = S(E) \begin{pmatrix} A \\ D \end{pmatrix}, \qquad S(E) = \begin{pmatrix} r_L & t_R \\ t_L & r_R \end{pmatrix}.} \tag{27}$$

The subscripts label the side of incidence. The two columns are determined by two independent scattering problems:

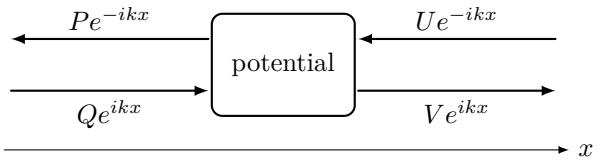
$$\begin{pmatrix} A \\ D \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \qquad \Longrightarrow \qquad \begin{pmatrix} B \\ C \end{pmatrix} = \begin{pmatrix} r_L \\ t_L \end{pmatrix}, \tag{28}$$

$$\begin{pmatrix} A \\ D \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \qquad \Longrightarrow \qquad \begin{pmatrix} B \\ C \end{pmatrix} = \begin{pmatrix} t_R \\ r_R \end{pmatrix}. \tag{29}$$

For right incidence, the transmitted wave leaves on the left and the reflected wave leaves on the right. This explains the order t_R, r_R in the second column. Once the matrix is known, any coherent combination of incoming waves can be treated by matrix multiplication.

Questions.

Question 6: identify incoming and outgoing waves.



Which pair gives the incoming and outgoing amplitudes, ordered (left, right)?

A) $\mathbf{a}_{\text{in}} = (P, U)^T, \quad \mathbf{a}_{\text{out}} = (Q, V)^T$

B) $\mathbf{a}_{\text{in}} = (Q, V)^T, \quad \mathbf{a}_{\text{out}} = (P, U)^T$

C) $\mathbf{a}_{\text{in}} = (Q, U)^T, \quad \mathbf{a}_{\text{out}} = (P, V)^T$

D) $\mathbf{a}_{\text{in}} = (P, V)^T, \quad \mathbf{a}_{\text{out}} = (Q, U)^T$

Answer and explanation. Answer: C. On the left, the right-moving wave Qe^{ikx} approaches the potential, while Pe^{-ikx} leaves it. On the right, the left-moving wave Ue^{-ikx} approaches and Ve^{ikx} leaves. The same factor e^{-ikx} is outgoing on the left but incoming on the right.

4.3 Coherent input from both sides

Reading the two rows of Eq. (27) gives

$$\boxed{B = r_L A + t_R D, \qquad C = t_L A + r_R D.} \tag{30}$$

Each outgoing amplitude can receive contributions from both inputs. If the incoming amplitudes have a controlled relative phase,

$$A = 1, \qquad D = e^{i\theta},$$

then the right-side outgoing amplitude is

$$C = t_L + r_R e^{i\theta}. \tag{31}$$

Its squared modulus contains an interference term:

$$|C|^2 = |t_L|^2 + |r_R|^2 + 2\,\mathrm{Re}\big(t_L^* r_R e^{i\theta}\big)\,. \tag{32}$$

Thus changing the relative phase can redistribute the outgoing flux. Here $|C|^2$ measures the right outgoing current in units of $\hbar k/m$. For these two unit-amplitude inputs, a fraction of the *total* incoming current would use the denominator $|A|^2 + |D|^2 = 2$.

Questions.

Question 7: two-sided input.
For $A = 1,\ D = e^{i\theta}$, the right-side outgoing amplitude C is:

A) $t_L + r_R e^{i\theta}$

B) $r_L + t_R e^{i\theta}$

C) $t_R + r_L e^{i\theta}$

D) $|t_L|^2 + |r_R|^2$

Answer and explanation. Answer: A. The second row gives $C = t_L A + r_R D$. Transmission of the left input and reflection of the right input contribute to the same outgoing channel, so their complex amplitudes must be added coherently.

Lecture 18 summary

- With the exterior potential set to zero, a finite well has discrete bound states at $-V_0 < E < 0$ and scattering states at $E > 0$.
- Positive-energy transmission is

$$T(E) = \left[1 + \frac{V_0^2}{4E(E+V_0)} \sin^2(k_2 L)\right]^{-1}, \qquad k_2 = \frac{\sqrt{2m(E+V_0)}}{\hbar}.$$

  The resonance condition is $k_2 L = n\pi$, giving $T = 1$ and $R = 0$.
- Every internal round trip adds the amplitude factor $r_{21} r_{23} e^{2ik_2 L}$. Summing all paths before taking the modulus squared explains the resonance.
- The S-matrix maps incoming to outgoing amplitudes:

$$\begin{pmatrix} B \\ C \end{pmatrix} = \begin{pmatrix} r_L & t_R \\ t_L & r_R \end{pmatrix} \begin{pmatrix} A \\ D \end{pmatrix}.$$

  Its first column describes left incidence and its second right incidence.
- For two-sided input, $C = t_L A + r_R D$: the superposition principle applies to amplitudes.

Next lecture. We will derive the constraints that probability current conservation places on S, relate bound states to its poles, and then turn to symmetry and angular momentum.

Suggested reading. Griffiths and Schroeter, *Introduction to Quantum Mechanics*, 3rd ed., the discussion of one-dimensional scattering from finite square wells and barriers. Review boundary matching, probability-current ratios, and transmission resonances. Use the lecture's channel ordering when working with the S-matrix.